

AdamW vs Muon for fine-tuning a small LM on SST-2

Setup. SmolLM2-135M on SST-2, linear head on last token. Muon (1e-3) on the 210 block matrices + AdamW (2e-5) on embedding/norms/head, vs AdamW (2e-5) on all 273 parameters. 1 epoch, bs 64, constant LR, wd 0, same data order, main pair 3 seeds, single-seed ablations on LR (both), momentum, weight decay.

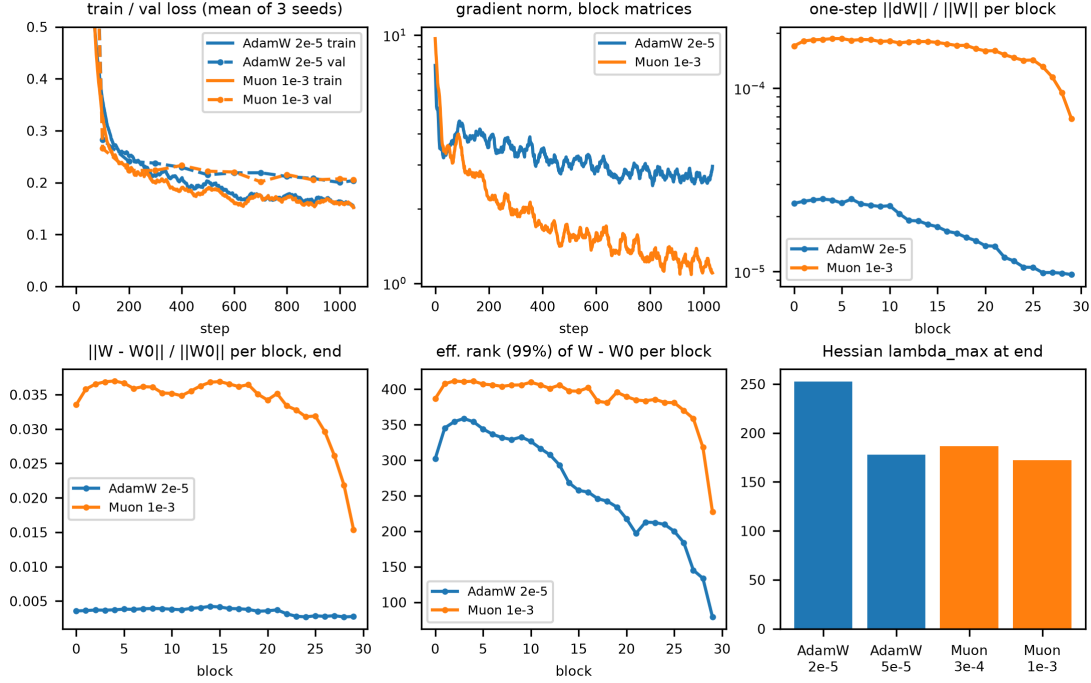


Figure 1: main pair, mean of 3 seeds. (a) train and val loss, (b) gradient norm of the block matrices, (c) one-step relative update per block, (d) relative drift from the pretrained weights per block, end of training, (e) effective rank of $W - W_0$ per block, (f) $\lambda_{\max}$ at the end of training.

Metrics

Chosen to separate *results/outcome* from *trajectory*. Outcome: val loss/acc, train loss. Trajectory: gradient norm of the block matrices, one-step $\|\Delta W\|/\|W\|$ and its stable rank (what the update rule does to a gradient), drift $\|W - W_0\|/\|W_0\|$ and the rank of $W - W_0$ (distance and directions moved from the checkpoint), Hessian $\lambda_{\max}$ by power iteration, with $L(W + \rho g/\|g\|) - L(W)$, $\rho = 0.05$, as a check.

config (seeds)	best acc	final acc	val loss	drift	$\lambda_{\max}$
AdamW 2e-5 (3)	.935±.002	.929±.002	.204	.0035	253±20
Muon 1e-3 (3)	.930±.005	.920±.004	.205	.034	172±33
AdamW 1e-5 (1)	.923	.921	.225	.0024	347
AdamW 5e-5 (2)	.935±.006	.929±.005	.191	.0069	178±5
Muon 3e-4 (2)	.932±.002	.931±.004	.200	.0107	187±7
Muon 3e-3 (2)	.909	.890	.269	.098	134±39
Muon 1e-3, $\mu=0.9$ (1)	.934	.919	.195	.025	208
Muon 1e-3, $\mu=0$ (1)	.919	.916	.337	.008	283
AdamW 2e-5, wd .01 (1)	.937	.929	.204	.0035	249
Muon 1e-3, wd .01 (1)	.936	.926	.195	.035	175

Table 1: Val metrics after 1 epoch (872 examples, SE ≈ 0.009), drift averaged over blocks, $\lambda_{\max}$ on 2 fixed batches.

Summary and mechanisms for the solutions

Muon’s step is orthogonalized, so one update has stable rank ~ 206 (of 576) vs ~ 3.5 for AdamW, and the accumulated change $W - W_0$ has effective rank 387 vs 265 (Fig. 1e). Muon’s step size is the same for every block and every step (Fig. 1c), so its drift scales linearly with LR (0.011, 0.034, 0.098 at 3e-4, 1e-3, 3e-3), AdamW’s drift grows sublinearly (0.0024, 0.0035, 0.0069, Fig. 1d). Muon’s gradient norm on the block matrices falls to $\sim 3\times$ below AdamW’s by the end of training (Fig. 1b), consistent with its lower train loss and flatter solution. Momentum is what makes the orthogonalized step full-rank: stable rank 72 / 184 / 206 at $\mu = 0 / 0.9 / 0.95$. Weight decay 0.01 changes accuracy by $\leq 0.3\%$ for both (within SE) and leaves Muon’s drift and $\lambda_{\max}$ essentially unchanged.

Which optimizer performed better, by what criterion?

On accuracy, AdamW, but only slightly: 0.935 ± 0.002 vs 0.930 ± 0.005 is within the ≈ 0.009 SE of 872 examples, and final

val loss is identical. On fit, Muon: lower train loss from step ~ 200 at equal val loss (Fig. 1a). On speed, AdamW at the conventional 2e-5 looks slower than Muon 1e-3, but at its best LR (5e-5) it reaches convergence at the same step. At the end of training, AdamW is more stable: under constant LR, Muon 1e-3 peaks at 0.930 and then drops to 0.920 in all three seeds, while AdamW remains stable. At Muon 3e-4 the drop is gone and every metric matches AdamW, with $3\times$ the drift instead of $10\times$.

Which solution appears flatter?

At default LR, Muon’s: $\lambda_{\max} = 172\pm 33$ vs 253 ± 20 (Fig. 1f), and the $\rho=0.05$ uphill loss increase agrees (0.06–0.10 vs 0.08–0.12).¹ But sharpness tracks step size, not optimizer: AdamW’s $\lambda_{\max}$ falls monotonically with LR (347, 253, 178), and the LR-matched pair AdamW 5e-5 / Muon 3e-4 land at 178 ± 5 / 187 ± 7 .

How reliable are these conclusions?

They hold for one narrow setting: a 135M model, a near-saturated classification task, full fine-tuning from an Adam-pretrained checkpoint at constant LR. Qu et al. (2026) report Muon matching or trailing Adam in exactly this regime, so a near-tie is expected; nothing here transfers to pretraining or RL. Within the setting: the accuracy tie and the end-of-training drop are reliable (consistent across all 3 seeds, and the seed-to-seed variation is small), Muon being flatter at default LR is moderately reliable (~ 2 SE), the LR-matched sharpness comparison is based on two configs, and the momentum and weight-decay rows are single-seed.

What would you change in a larger study?

Broaden the setting first: a task without a ceiling (MNLI, or generative SFT on GSM8K), larger models, and RL. Momentum is essential in SFT where consecutive gradients align, but under sparse rewards gradient alignment is near zero (Mukherjee et al.), so Muon’s behaviour may change. A Muon-pretrained checkpoint, to test the mismatch explanation. Then LR decay, more seeds per LR, and more runs with tuned weight-decay values.

¹Spectral norms of W_q, W_k are unchanged (ratio 1.000) in every run, so this is not a weight-scale artifact (Dinh et al. 2017).